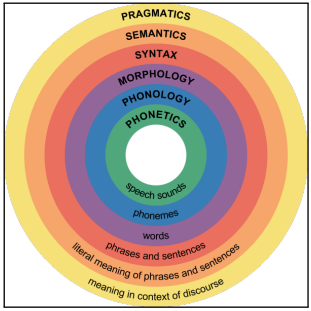
Levels of linguistic structure

- phonology: sounds
- morphology: structure of words
- syntax: structure of sentences
- semantics: the meaning of words, and how they combine to form larger meanings
- pragmatics: the meaning of utterances within a discourse context



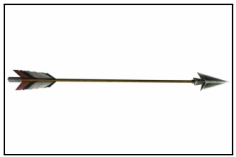
Ambiguity: phonology

- homophones : buy, by, bye
- important for speech recognition



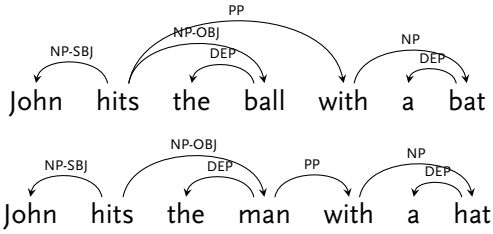
Ambiguity: morphology

- homonyms/homographs:
  - flies (noun) / flies (verb)
  - past (preposition) / past (noun)
  - like (preposition) / like (verb)
- complicates syntactic analysis
- complicates semantic analysis
- when combined, the ambiguity becomes even more important
  - Time flies like an arrow.
  - Fruit flies like a banana.



Ambiguity: syntax

Structural ambiguity







Ambiguity: semantics

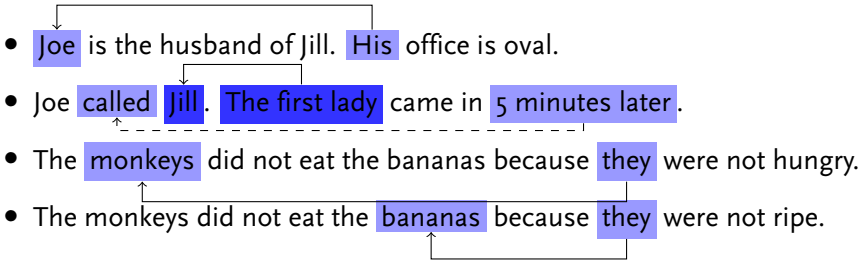
- Leslie used a *pen* to write this letter.
- We put he pigs in a *pen*.
- The judge sent them to the *pen* for a decade.
- They should *pen* that letter quickly.







Ambiguity: pragmatics



Difficulties

Machine translation

- word order in English is *modifier-noun*
  word order in French is *noun-modifier*
- English: *a red house*
  French: *une maison rouge*
- English: *almost double the usual oatmeal biscuits*
  French: *presque le double des biscuits à l'avoine habituels*





Difficulties

Machine translation

- Word can have different meanings
- e.g. French *avocat*

Difficulties

Machine translation

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Difficulties

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Difficulties

Machine translation

- Word can have different meanings
- e.g. French *avocat*



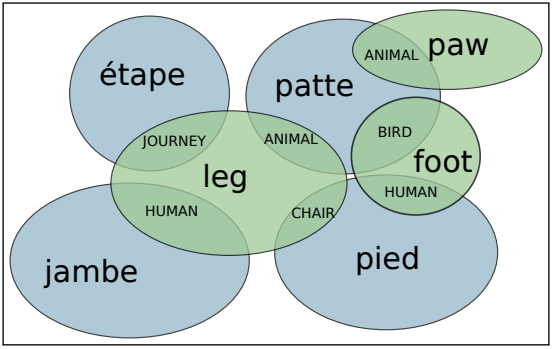
avocado



lawyer

Difficulties

Machine translation



Difficulties

Machine translation

- Certain languages, like Spanish, can omit subject pronouns
  - –o = I
  - –as = you
  - –a = he/she/it
  - –amos = we
  - –an = they
- When should the system use *he, she, it*?



Difficulties

Machine translation

- The monkeys did not eat the bananas because *they* were not hungry.
- Les singes n'ont pas mangé les bananes parce qu'*ils* n'avaient pas faim.
- The monkeys did not eat the bananas because *they* were not ripe.
- Les singes n'ont pas mangé les bananes parce qu'*elles* n'étaient pas mûres.



Difficulties

Machine translation

- I saw *her* yesterday.
- Ich habe *sie* gestern gesehen.
- I gave *her* a present.
- Ich habe *ihr* ein Geschenk gegeben.
- I also gave *her* brother a present.
- Ich habe auch *ihrem* Bruder ein Geschenk gegeben.



Difficulties

Machine translation

- French: *je bois du lait*



Difficulties

Machine translation

- French: *je bois du lait*
- English: *I drink milk* ?



Difficulties

Machine translation

- French: *je bois du lait*
- English: *I drink milk* ?
- English: *I'm drinking milk* ?



Difficulties

Machine translation

- *bone of contention*
- *kick the bucket*
- *Don't sell the bear's fur before hunting it.*







Difficulties

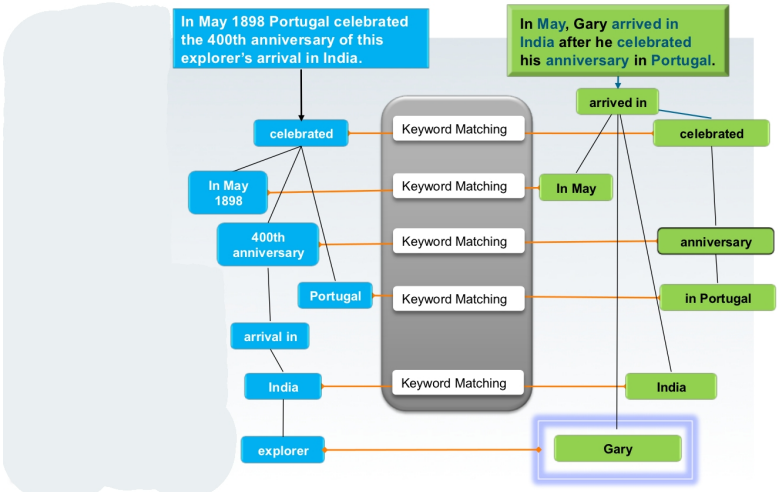
Spell-checking

- * This is too hot too drink.
- * This is to hot to drink.
- * This is to hot too drink.
- This is too hot to drink.



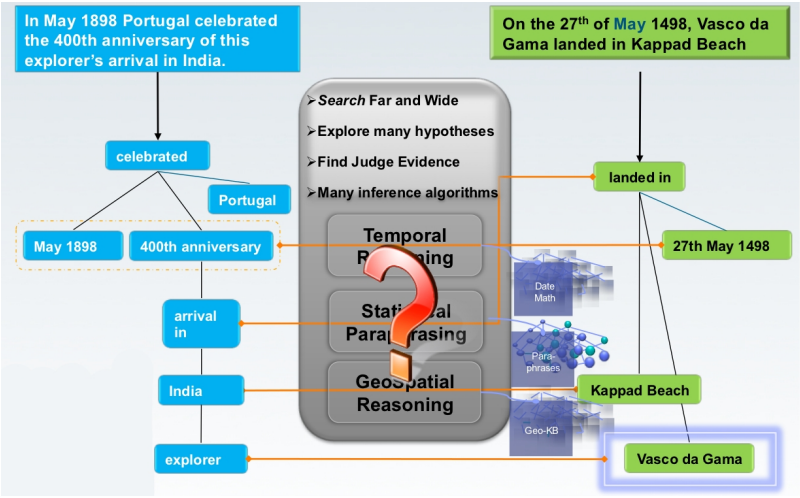
Difficulties

Question-Answering



Difficulties

Question-Answering



Difficulties

Speech technology

- Speech Recognition
  - fish vs. ghoti
  - The system is trained to wreck a nice peach
- Speech Synthesis
  - conténт (adjective) vs. cóntent (noun)
  - addréss (verb) vs. áddress (noun)







Difficulties

Speech technology

- Speech Recognition
  - fish vs. ghoti rough women nation
  - The system is trained to wreck a nice peach
- Speech Synthesis
  - conténts (adjective) vs. cóntent (noun)
  - addréss (verb) vs. áddress (noun)







Difficulties

Speech technology

- Speech Recognition
  - fish vs. ghoti rough women nation
  - The system is trained to recognize speech
- Speech Synthesis
  - conténts (adjective) vs. cóntent (noun)
  - addréss (verb) vs. áddress (noun)







FACULTY OF ARTS

KU LEUVEN

Computational Linguistics

Lecture 2: Probability and machine learning

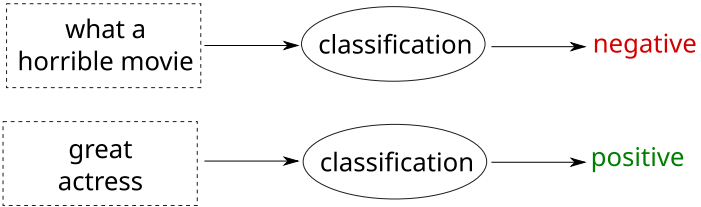
Tim Van de Cruys



Tuesday 29 September 2026

Classification

- **Classification** = automatically assigning an instance to one of a predefined set of categories
- Widely used in NLP:
  - Document/topic classification (e.g. sports, politics, culture)
  - Sentiment analysis (positive, negative, neutral)
  - Part-of-speech tagging (noun, verb, adjective, ...)
  - Named entity recognition (person, location, organization, ...)
- Often a binary task (yes/no, true/false), but can also be multi-class or multi-label



Rule-based

1. IF exists word in document
   such that word in `['exciting', 'entertaining', ...]`
   THEN output POSITIVE
2. IF exists word in document
   such that word in `['hated', 'worst', ...]`
   THEN output NEGATIVE

- Easy to interpret and adjust
- But: labor-intensive and not generalizable

Rule-based

1. IF exists word in document
   such that word in `['soccer', 'tennis', ...]`
   THEN output **SPORTS**
2. IF exists word in document
   such that word in `['rainy', temperature', ...]`
   THEN output **WEATHER**

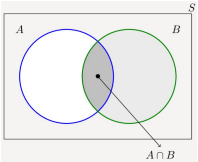
- Easy to interpret and adjust
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Supervised learning: statistics

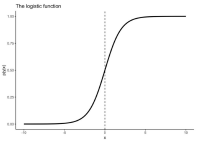
- Data-driven approach
- Computer determines on its own what is the importance of features
- Input:
  - Set of labels (classes): { +, − }
  - Set of labeled documents: { (d_1,+), (d_2,−),(d_3, +), ...}
- Output:
  - Trained classifier: $F : d \rightarrow c$

- What form does *F* take?
- How can we learn *F*?

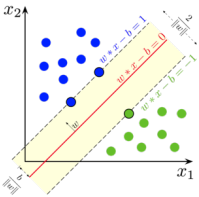
Supervised classifiers



Naive Bayes



logistic regression



Support vector machines



K-nearest neighbours

Machine learning

supervised and non-supervised

- Supervised learning
  - Train a system on a classification task
  - We need a **training set**: a set of manually labeled instances
  - We equally define important characteristics: **features**
  - A machine learning algorithm allows us to associate features with labels
  - Performance is measured on a separate **test set**
- Unsupervised learning
  - We induce object representations automatically, in order to generalize without manual intervention

Supervised learning: feature-based learning

- Supervised learning is based on the concept of **features** (descriptors, characteristics)
- Elementary indices, that link certain aspects of the observed data (d) to the class that we want to predict (c)
- The prediction of the class c depends exclusively on the features f, constructed on the basis of the data d
- Each feature has a numerical/categorical/boolean value
- We apply machine learning algorithms on the basis of these features

Feature model

Examples

- Text categorization
  - Class *c*: 'medical'
  - Data *d*: document
  - Features *f*: words (boolean/word count)
- Author identification
  - Class *c*: 'Charles Dickens'
  - Data *d*: document (book)
  - Features *f*: extracted n-grams
- Sentiment classification
  - Class *c*: 'negative sentiment'
  - Data *d*: tweet
  - Features *f*: words

Feature-based model

Linear classifier

- Classification from a set of features $\{f_i\}$ to a number of classes $\{c\}$
- Attribute a weight $\lambda_{i,c}$ for each feature f_i and each class c
- For a certain pair (c, d), the features will vote with their weights
- We pick the class c that maximizes the sum of the various features
- Different ways for choosing the weights (optimization)

Feature-based model

Linear classifier: example

- Text classification
- As features, we take the frequency of lemmas
- $\max_{c \in C} \sum_i \lambda_{i,c} f_i$

"The virus predominantly targets Windows computers."

	$\lambda_{i,j}$	
	class c_j	
feature f_i	computer science	biology
virus	0.5	0.6
target	0.4	0.2
computer	0.8	-0.6
Windows	0.9	-1.2

- $\mathit{vote}(\mathit{computer\ science}) = 0.5 \times 1 + 0.4 \times 1 + 0.8 \times 1 + 0.9 \times 1 = 2.6$
- $\mathit{vote}(\mathit{biology}) = 0.6 \times 1 + 0.2 \times 1 + (-0.6) \times 1 + (-1.2) \times 1 = -1.0$

Feature-based model

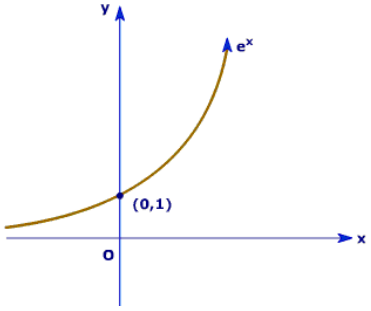
Logistic regression

- Linear scores (range $-\infty$ to ∞) are difficult to interpret, we'd like to have probabilities
- We construct a probabilistic model on the basis of the linear model
- Transform the values to probabilities using the exponential function + normalization
- We choose the parameters $\lambda_{c,i}$ that maximize the likelihood based on the training data

$$P(c \mid d, \lambda) = \frac{\exp(\sum_i \lambda_{c,i} f_i)}{\sum_{c' \in C} \exp(\sum_i \lambda_{c',i} f_i)}$$

- $P(\textit{computer science} \mid d) = e^{2.6}/(e^{2.6} + e^{-1.0}) = 0.97$
- $P(\textit{biology} \mid d) = e^{-1.0}/(e^{2.6} + e^{-1.0}) = 0.03$

Exponential function



Sentiment analysis

- Monitor the opinion about a product
- Aggregate opinions
- Monitor trends

Sentiment analysis

- Monitor the opinion about a product
- Aggregate opinions
- Monitor trends



Sentiment analysis

- Monitor the opinion about a product
- Aggregate opinions
- Monitor trends



vs.

35 out of 173 people found the following review useful:

Another awful overrated film

Author: Wormhole10

13 April 2016

Terrible, overrated and plain boring. It's obvious the critics were paid shills this movie was just awful. Oh where to start.. First off all the characters. Like all Disney made marvel movies the characters lack any kind of development. There is no attachment to them and the audience can care less what happens to them. The soundtrack is just another generic computer generated sound and lacks any real soul. The plot is also a rushed mess. Moreover this movie gives the same generic marvel formula that we have come to expect. Batman VS Superman was a much better film and Disney should learn from them and take lessons. If you want a true Comic Book film save your time and go watch Batman V Superman, if you want more mindless explosions,kiddie jokes and characters without depth this is the film for you.

Was the above review useful to you? Yes No

24 out of 232 people found the following review useful:

The Worst Marvel Movie and One of the Worst Action Movies Ever Made

Author: AndreCedeno420 from United States

13 April 2016

Awful, just awful. This boring, overlong 2 hour and 20 minute flick is executed with terrible camera work, bad acting, terrible CGI, and cringe worthy dialogue. All the characters are underdeveloped and weak, you don't care for any of them. The score is completely awful. This is another typical Marvel movie that continues to the lower the standards for Marvel films. This movie sucks like every other Marvel film, DC is better and always the best. Everything about this movie is boring and no one should see it. I suggest you people go see a real movie instead, it's called Batman V Superman. Everything about this movie is terrible and it is way more worth your money to go see a masterpiece like Batman V Superman instead.

0/10

Was the above review useful to you? Yes No

Sentiment analysis

Example: movie review

- 👍 It was an exciting, entertaining film.
- 👎 I'm a massive bond fan and I can't begin to describe how much I hated this movie.
- 👎 One of the worst films I have ever seen. Every single scene was flawed and had errors. From beginning to end the movie had mistake after mistake.
- 👍 This is one of the best Bond movies I have ever seen. The story is superbly put together and has some interesting twists, the action is well done and contains none of the shaky cam which plagued the last film.
- 👍 It's the Bond film we've all been waiting for!

Sentiment analysis

Different degrees of complexity

- Simple
  - Is the attitude of this document positive or negative?
- Moderate
  - Sentiment on a scale from 1 to 5
- Advanced
  - Detect the target, source, complex type of attitude

Sentiment analysis

Base algorithm

- Tokenization (what is a word? $\rightarrow$ next lecture)
- Feature extraction
- Classification
  - Naive Bayes
  - Logistic regression
  - Support vector machines

Naive Bayes classification

- Simple classification ('naive'), based on Bayes' theorem
- Applied to a simple document representation (bag of words)

Bag of words for sentiment analysis

$\gamma($ `This is one of the best Bond movies I have ever seen. The story is superbly put together and has some interesting twists. The actors all do a great job.` $) = c$



Bag of words for sentiment analysis

$\gamma($ This is one of the **best** Bond movies I have ever seen. The story is **superbly** put together and has some **interesting** twists. The actors all do a **great** job. $) = c$



Bag of words for sentiment analysis

$$\gamma\Big($$ xxxx xx xxx xx xxx **best** xxxx xxxxx x
xxxx xxxx xxxxx xxx xxxxx xx **superbly** xxx
xxxxxxxx xxx xxx xxxx **interesting twists** xxx
xxx xxxxxx xxx xx x **great** xxxx $$\Big) = c$$



Bag of words for sentiment analysis

$\gamma($

best	1
superbly	1
interesting	1
twists	1
great	1

$) = c$



Bayes' theorem for sentiment classification

- For document d and class c (positive/negative)

$$P(c \mid d) = \frac{P(d \mid c)P(c)}{P(d)}$$

Bayes' theorem for sentiment classification

$$
\begin{aligned}
c_{MAP} &= \underset{c \in C}{\operatorname{argmax}}\, P(c \mid d) \\
&= \underset{c \in C}{\operatorname{argmax}}\, \frac{P(d \mid c)P(c)}{P(d)} \\
&\simeq \underset{c \in C}{\operatorname{argmax}}\, P(d \mid c)P(c) \\
&\simeq \underset{c \in C}{\operatorname{argmax}}\, P(x_1, x_2, \ldots, x_n \mid c)P(c)
\end{aligned}
$$

Bayes' theorem for sentiment classification

$$c_{MAP} = \underset{c \in C}{\operatorname{argmax}}\, P(x_1, x_2, \ldots, x_n \mid c) P(c)$$

- What is the probability of a class given that **all features appear together in a document**
- Impossible to estimate this probability, unless we have a gigantic training set

Independence hypotheses

- Bag of words hypothesis: hypothesis that the order of words is not important
- Conditional independence hypothesis: hypothesis that the probability of features (given the class) is independent
  - $P(x_1, x_2, \ldots, x_n \mid c) = P(x_1 \mid c) \cdot P(x_2 \mid c) \cdot \ldots \cdot P(x_n \mid c)$

Naive Bayes classifier

$$c_{MAP} = \operatorname*{argmax}_{c \in C} P(x_1, x_2, \ldots, x_n \mid c) P(c)$$

$$c_{NB} = \operatorname*{argmax}_{c \in C} P(c) \prod_{x_i \in X} P(x_i \mid c)$$

Train a naive Bayes classifier

- First try: estimate maximum likelihood
- We use the frequencies of the data

$$P(c_j) = \frac{doccount(C = c_j)}{N_{doc}}$$

$$P(w_i \mid c_j) = \frac{count(w_i, c_j)}{\sum_{w \in V} count(w, c_j)}$$

Train a naive Bayes classifier

$$P(w_i \mid c_j) = \frac{count(w_i, c_j)}{\sum_{w \in V} count(w, c_j)}$$

- Fraction of time we see word w_i out of all the words in the documents with class c_j
- How to estimate this quantity efficiently?
  - We create a large meta-document for a class j by concatenating all the documents of the same class
  - We use the frequency of w in this document

Problem with maximum likelihood estimation

- What happens when we don't have any training documents with the word *fantastic*, labeled with the class *positive* ?

$$P(\textit{fantastic} \mid \textit{positive}) = \frac{\textit{count}(\textit{fantastic}, \textit{positive})}{\sum_{w \in V} \textit{count}(w, \textit{positive})} = 0$$

- Problem: if a single probability in our product is zero, the entire probability is zero (even if we have other good evidence)
- Zeros have a destructive influence

$$c_{NB} = \underset{c \in C}{\operatorname{argmax}}\, P(c) \prod_{x_i \in X} P(x_i \mid c)$$

Laplace smoothing (add-1) for naive Bayes classification

$$P(w_i \mid c_j) = \frac{count(w_i, c_j) + 1}{\sum_{w \in V}(count(w, c_j) + 1)}$$
$$= \frac{count(w_i, c_j) + 1}{(\sum_{w \in V} count(w, c_j)) + |V|}$$

Naive bayes classifier: algorithm

1. Extract the vocabulary V from our training corpus
2. Compute the probabilities $P(c_j)$
   - for each class c_j in C do
     - $docs_j \leftarrow$ all docs with class $= c_j$
     - $P(c_j) \leftarrow \frac{|docs_j|}{|\#documents|}$
3. Compute the probabilities $P(w_i \mid c_j)$
   - $text_j \leftarrow$ concatenation of all $docs_j$
   - for each word w_i in V do
     - $n_i \leftarrow$ # occurrences of w_i in $text_j$
     - $P(w_i \mid c_j) \leftarrow \frac{n_i+1}{n+|V|}$

Other classifiers for sentiment classification

- Naive bayes is not the only possibility
- Other approaches:
  - Logistic regression
  - Support vector machines (SVM)
  - Neural networks
- Good implementations of various classifiers already exist; one just needs to define the features, and provide the training data

Sentiment classification: difficulties

- Sentiment classification can be difficult
  - Subtlety
    - “she expresses the entire range of emotions, from A to B”
  - Different expectations
    - “this movie should have been fantastic”
  - Opinion on related expectation
    - “I am disappointed to read all these negative comments”

Structured prediction: sequences

- Many natural language processing tasks take sequences as input
- The task is then to label each element of the sequence

Charles	Michel	is	the	president	of	the
European	Council	since	december	2019	.	

Structured prediction: sequences

- Many natural language processing tasks take sequences as input
- The task is then to label each element of the sequence

POS tagging

Charles	Michel	is	the	president	of	the
NPP	NPP	V	DET	NC	P	D

European	Council	since	december	2019	.
ADJ	NPP	P	NC	NUM	PUNCT

Structured prediction: sequences

- Many natural language processing tasks take sequences as input
- The task is then to label each element of the sequence

Named entity recognition

Charles	Michel	is	the	president	of	the
NPP	NPP	V	DET	NC	P	D
B-PER	I-PER	O	O	O	O	O

European	Council	since	december	2019	.
ADJ	NPP	P	NC	NUM	PUNCT
B-ORG	I-ORG	O	B-DAT	I-DAT	O

Part of speech

- Words can be categorised in classes that behave similarly
  1. I can see a _____.
  2. The _____ is on the table.
  3. She's thinking about a _____.
- Part of speech provides information about the context in the sentence
  - *adjective* often succeeded by *noun*
  - *personal pronoun* often succeeded by *verb*
- Useful information for language processing applications
- Automatically assigning part of speech: **part of speech tagging**



Named entity recognition

- An important subtask in NLP
- Find and classify the names in a text

According to a recent survey published by Eurostat on Wednesday 28 October, the gap between the hourly cost of labour in Paris and in Berlin is tightening.

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Named entity recognition

- An important subtask in NLP
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According to a recent survey published by Eurostat on Wednesday 28 October, the gap between the hourly cost of labour in Paris and in Berlin is tightening.

- person
- location
- date
- organization

Named entity recognition

IOB Encoding

```
published   O
by          O
Eurostat    B-ORG
on          O
Wednesday   B-DAT
28          I-DAT
October     I-DAT
```

- Using a suitable encoding scheme, named entity recognition becomes a sequence labeling task
- **I**nside, **O**utside, **B**eginning
- B-XX in order to separate adjacent entities of the same type

Manual definition of features

- Words
  - Current word
  - Previous/next word (context)
  - Substring of characters
  - Word form
- Additional linguistic annotations
  - POS
- Context labels
  - Previous/next label

Paramedics	treat	woman	and	boy
after	crash	in	Stoke-on-Trent	.

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Paramedics	treat	woman	and	boy
after	crash	in	Xx-x-Xx	.

Manual definition of features

- Words
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Paramedics	treat	woman	and	boy
NNS	VBP	NN	CC	NN
after	crash	in	Stoke-on-Trent	.
IN	NN	IN	NNP	PUNCT

Manual definition of features

- Words
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  - POS
- Context labels
  - Previous/next label

Paramedics	treat	woman	and	boy
NNS	VBP	NN	CC	NN
O	O	O	O	O
after	crash	in	Stoke-on-Trent	.
IN	NN	IN	NNP	PUNCT
O	O	O	?	O

Manual definition of features

- Words
  - Current word
  - Previous/next word (context)
  - Substring of characters
  - Word form
- Additional linguistic annotations
  - POS
- Context labels
  - Previous/next label

Paramedics	treat	woman	and	boy
NNS	VBP	NN	CC	NN
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after	crash	in	Stoke-on-Trent	.
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Sequence tagging

- Labeling a word in a sequence is the same as a simple classification
- We have predicted labels for previous words
- We use the predicted labels of previous words, as well as observed features (based on current, previous, and future words) in order to predict the best possible label for the current word

crash	in	Stoke-on-Trent	.
NN	IN	NNP	PUNCT
O	O	?	?

Features	
w_0	Stoke-on-Trent
w_{-1}	in
w_{+1}	.
pos_0	NNP
pos_{-1}	IN
pos_{+1}	PUNCT
hasHyphen	True
$label_{-1}$	O

Global optimization

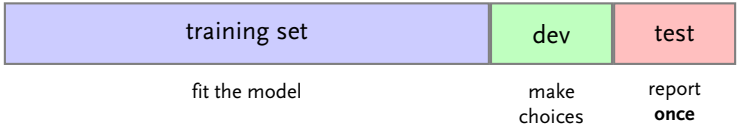
- Problem: we perform a series of local judgements, without taking into account the global optimality
- We hope that the local conditions give us a global optimum
- To solve our problem:
  - Instead of emitting a single answer, predict the *n* best answers
  - This way, we can compute the best global sequence

Experiment

- We have three sentiment classifiers
  - a list of hand-written **rules** (symbolic)
  - **naive Bayes** (statistical)
  - **logistic regression** (statistical)

- Which one is better? how can we find out?
- Common procedure in NLP: **experiment**

Train, development, test



- A model must be evaluated on data it has **never seen**: we care about generalization, not memorization
- **Training** set: estimate the parameters (probabilities, weights)
- **Development** (validation) set: for every choice you make (features, smoothing, thresholds, which model to keep: hyperparameters)
- **Test** set: touched once, at the very end, for the number you report
- Typical splits: 80/10/10 (train/dev/test)
- Little data? **Cross-validation** on the training part; the test set still stays apart

Leakage

- **Leakage**: information about the test set finds its way into the model, so the reported score is too optimistic
- It rarely looks like cheating:
  - you tried ten settings and kept the one that was best *on the test set*
  - the vocabulary was computed on *all* the data, before splitting
  - the same review (or a near-duplicate) occurs in both training and test
  - texts by the same author are included in both training and test
- Everything that is *learned* or *chosen* must be learned or chosen without looking at the test set

Baseline

- *'Our model reaches 95% accuracy'*: is that good?
- Example: toxic comment detection
  - about 95% of the comments are *not* toxic
  - A model that always answers `not toxic` scores 95%, while it does not find a single toxic comment
- A number only means something next to a **baseline**
  - majority class (always predict the most frequent label)
  - a handful of simple rules, a word list
  - the simplest model that could possibly work
- If your sophisticated model does not beat the simple baseline, you have a negative result

Upper bound

- Opposite of the baseline: the best score that can be expected
- Usually **human performance**: inter-annotator agreement
  - two annotators label the same items independently
  - percentage of agreement, or agreement corrected for chance (e.g. Krippendorff's α)
- The gold standard is only as reliable as the agreement between annotators
  - e.g. sentiment analysis: annotators often agree on 80–90% of the items
  - subjective tasks (irony, toxicity): considerably lower
- A score is interpreted between baseline and upper bound

Beyond accuracy

Confusion matrix

		gold standard	
		toxic	not toxic
system	toxic	true positive (TP)	false positive (FP)
	not toxic	false negative (FN)	true negative (TN)

$$P = \frac{TP}{TP + FP} \qquad R = \frac{TP}{TP + FN} \qquad F_1 = \frac{2PR}{P + R}$$

- The always-`not toxic` baseline: accuracy 95%, but recall and F_1 for the class *toxic*: 0%

Multiple classes

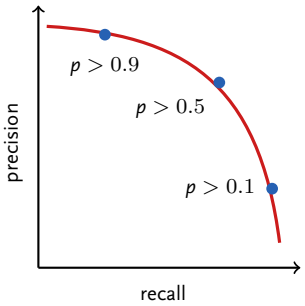
- With more than two classes: compute P, R, F_1 per class, then average
  - **micro**-average: pool all decisions $\rightarrow$ dominated by the frequent classes
  - **macro**-average: mean of the per-class scores $\rightarrow$ every class counts equally
- With imbalanced classes, both averages can differ considerably

Performance on sequences

- How to evaluate named entity recognition?
- Precision, recall and F-measure, computed over **entities** rather than words
  - precision: percentage of entities returned by the system that are correct
  - recall: percentage of the entities in the test set that are found by the system
- An entity is correct if both its boundaries and its type correspond to the test set
  - *Charles Michel* as PER, but only *Michel* tagged: incorrect
  - *European Council* tagged as LOC: incorrect
- Per-word accuracy would be misleadingly high, as most words are labeled O

Thresholds

- Naive Bayes and logistic regression output a **probability**
- A decision requires a **threshold**, e.g. toxic if $p > 0.5$
  - lower threshold: higher recall, lower precision
  - higher threshold: higher precision, lower recall
- The threshold is a hyperparameter: it is set on the development set



Significance

- Example: naive Bayes 84.2% accuracy, logistic regression 84.7% accuracy
- Is logistic regression better? Not necessarily
  - the test set is a sample: a different set of 1,000 reviews yields a different score
  - models with random components (initialization, order of the data) yield different results with different random seeds
- Guidelines:
  - use a sufficiently large test set; report size
  - repeat with different seeds or folds; report mean and standard deviation
  - small differences between models are often not significant: use a significance test (e.g. McNemar's test, paired bootstrap)

Error analysis

- An evaluation score indicates how often a model makes mistakes, not why
- Error analysis: inspection of the errors on the development set
  - take a sample of errors (e.g. 50) and categorize them
  - which categories are frequent, and which could be addressed?
- Linguistic knowledge is often helpful
- Goal: explain a pattern of errors

Checklist

1. Keep a separate test set, and use it only once
2. Avoid leakage
3. Compare against a baseline
4. Choose metrics that suit the task and the class distribution
5. Make all choices (incl. thresholds) on the development set
6. Report the uncertainty of the results
7. Analyze the errors
8. Support every claim with results